

$$\textcircled{1} \quad \tan x \frac{\partial z}{\partial x} + \tan y \frac{\partial z}{\partial y} = \tan z.$$

$$AE = \frac{dx}{\tan x} = \frac{dy}{\tan y} = \frac{dz}{\tan z}.$$

$$\frac{dx}{\tan x} = \frac{dy}{\tan y} \Rightarrow \int \frac{dx}{\tan x} = \int \frac{dy}{\tan y}$$

$$\Rightarrow \int \cot x \, dx = \int \cot y \, dy$$

$$\Rightarrow \log(\sin x) = \log(\sin y) + \log c_1$$

$$\Rightarrow \frac{\sin x}{\sin y} = c_1.$$

$$\frac{dx}{\tan x} = \frac{dz}{\tan z} \Rightarrow \frac{\sin x}{\sin z} = c_2.$$

$$\therefore \text{Solution is } f\left(\frac{\sin x}{\sin y}, \frac{\sin x}{\sin z}\right) = 0.$$

$$3. \quad x^2(y-z)p + y^2(z-x)q = z^2(x-y)$$

$$\frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)}$$

$$P = x^2(y-z) \quad Q = y^2(z-x) \quad R = z^2(x-y)$$

$$\int \frac{1}{x^2} dx = \int x^{-2} dx = \frac{x^{-2+1}}{-2+1} = -\frac{1}{x}$$

Choose l, m, n s. $lP + mQ + nR = 0$.

$$\text{Let } l = \frac{1}{x^2}, m = \frac{1}{y^2}, n = \frac{1}{z^2} \text{ then}$$

$$\frac{1}{x^2} [x^2(y-z)] + \frac{1}{y^2} [y^2(z-x)] + \frac{1}{z^2} [z^2(x-y)] = 0$$

$$\begin{aligned} \therefore \frac{1}{x^2} dx + \frac{1}{y^2} dy + \frac{1}{z^2} dz &= 0 \\ \Rightarrow \frac{-1}{x} - \frac{1}{y} - \frac{1}{z} &= -\frac{1}{c} \\ \Rightarrow \frac{1}{x} + \frac{1}{y} + \frac{1}{z} &= \frac{1}{c} \end{aligned}$$

$$\text{Next let } l = yz, m = xz, n = xy$$

$$\therefore lP + mQ + nR = 0$$

$$\Rightarrow (yz)dx + (xz)dy + (xy)dz = 0$$

$$\Rightarrow d(xyz) = 0$$

$$\Rightarrow xyz = c_2$$

$$\begin{aligned} & x^2y - x^2z + y^2z - y^2x + z^2x - z^2y \\ & yz(x^2y - x^2z) = x^2y^2z - x^2yz^2 \\ & xyz(y^2z - y^2x) = xy^2z^2 - x^2y^2z \\ & xyz(z^2x - z^2y) = x^2yz^2 - xy^2z^2 \end{aligned}$$

$\therefore$ Solution is $f(xyz, \frac{1}{x} + \frac{1}{y} + \frac{1}{z}) = 0$.

OR : $(l, m, n) = (\frac{1}{x}, \frac{1}{y}, \frac{1}{z})$

1. Solve $(mz - ny) \frac{dz}{dy} + (nx - lz) \frac{dz}{dy} = ly - mx$.

** $P = mz - ny$, $Q = nx - lz$, $R = ly - mx$.

$$\frac{dx}{mz - ny} = \frac{dy}{nx - lz} = \frac{dz}{ly - mx}$$

Choose l', m', n' s.t. $l'P + m'Q + n'R = 0$.

$$l(mz - ny) + m(nx - lz) + n(ly - mx) = 0$$

$$\therefore ldx + mdy + ndz = 0$$

$$\Rightarrow lx + my + nz = C_1$$

Now choose multipliers as x, y, z .

then $x(mz - ny) + y(nx - lz) + z(ly - mx) = 0$

$$\therefore xdx + ydy + zdz = 0$$

$$\Rightarrow x^2 + y^2 + z^2 = C_2$$

$\therefore$ Solution is $f(lx + my + nz, x^2 + y^2 + z^2) = 0$

Solve $(3z - 4y)p + (4x - 2z)q = 2y - 3z$.

Solve $(2z - y)p + (x + z)q = -2x - y$.

$P = 2z - y, Q = x + z, R = -2x - y$.

$$\frac{dx}{2z - y} = \frac{dy}{x + z} = \frac{dz}{-2x - y}$$

Choose l, m, n s.t. $lP + mQ + nR = 0$.

$l = 1, m = 2, n = 1$

$$dx - 2dy - dz = 0$$

$$\Rightarrow x - 2y - z = c_1$$

Choose $l = x, m = y, n = z$, so that

$$x(2z - y) + y(x + z) + z(-2x - y) = 0$$

$$\therefore xdx + ydy + zdz = 0$$

$$\Rightarrow x^2 + y^2 + z^2 = c_2$$

Solution is $(x - 2y - z, x^2 + y^2 + z^2) = 0$.

7. Solve $(y+zx)p - (x+yz)q = x^2 - y^2$.

$P = y+zx$, $Q = -(x+yz)$, $R = x^2 - y^2$

AE: $\frac{dx}{y+zx} = \frac{dy}{-(x+yz)} = \frac{dz}{x^2 - y^2}$

Choose $l = x$, $m = y$, $n = -z$

then $x(y+zx) - y(x+yz) - z(x^2 - y^2) = 0$

$\Rightarrow xy + zx^2 - xy - y^2z - zx^2 + zy^2 = 0$

$\therefore xdx + ydy - zdz = 0 \Rightarrow x^2 + y^2 - z^2 = C_1$

$\frac{dx + dy}{y + zx - x - yz} = \frac{dz}{x^2 - y^2}$

$\Rightarrow \frac{dx + dy}{y - x + z(x - y)} = \frac{dz}{x^2 - y^2}$

$\Rightarrow \frac{dx + dy}{z(x - y) - (x - y)} = \frac{dz}{(x - y)(x + y)}$

$\Rightarrow \frac{dx + dy}{(x - y)(z - 1)} = \frac{dz}{(x - y)(x + y)}$

$\Rightarrow (x + y)(dx + dy) = (z - 1)dz$

$y + zx - x - yz = \frac{y^2 - x^2}{x^2 - y^2}$

$y(1 - z) = y - x + z(y - x)$

$x + y = u$

$dx + dy = du$

$u du = (z - 1) dz$

$\frac{u^2}{2} = \frac{z^2}{2} - z + C$

$(x + y)^2 - z^2 - 2z = C$

$x dx + y dx + x dy + y dy$

$$\Rightarrow xdx + xdy + ydx + ydy = (z-1)dz.$$

$$\Rightarrow xdx + (xdy + ydx) + ydy = (z-1)dz.$$

$$\Rightarrow \frac{x^2}{2} + xy + \frac{y^2}{2} = \frac{z^2}{2} - z + C$$

$$\Rightarrow (x+y)^2 = z^2 - 2z + C_2.$$

$$\Rightarrow (x+y)^2 - z^2 + 2z = C_2.$$

OR

Choose $l = y, m = x, n = 1$

$$y(y + zx) - x(x + yz) + x^2 - y^2 = 0.$$

$$\therefore ydx + xdy + dz = 0.$$

$$\Rightarrow xy + z = C_2.$$

$$\therefore \text{Soln. is } \underline{\underline{f(xy + z, x^2 + y^2 - z^2) = 0.}}$$

8. Solve $z(x+y)p + z(x-y)q = x^2 + y^2$.

$P = z(x+y), Q = z(x-y), R = x^2 + y^2$

AE: $\frac{dx}{z(x+y)} = \frac{dy}{z(x-y)} = \frac{dz}{x^2 + y^2}$.

Choose $l = x, m = -y, n = -z$.

$$\begin{aligned} x(zx + zy) - y(zx - zy) + z(x^2 + y^2) \\ = zx^2 + zxy - zxy + zy^2 - zx^2 - zy^2 \\ = 0 \end{aligned}$$

$$\therefore xdx - ydy - zdz = 0$$

$$\Rightarrow x^2 - y^2 - z^2 = c_1$$

$$\frac{dx}{z(x+y)} = \frac{dy}{z(x-y)} \Rightarrow (x-y)dx = (x+y)dy$$

$$\Rightarrow xdx - ydx = xdy + ydy$$

$$\Rightarrow xdx - ydy = xdy + ydx$$

$$\Rightarrow xdx - ydy = d(xy)$$

$$\Rightarrow x^2 - y^2 - 2xy = c_2$$

$\therefore$ Solution is $f(x^2 - y^2 - 2xy, x^2 - y^2 - z^2) = 0$.

9. Solve $(z^2 - 2yz - y^2) + (xy + zx)q = xy - zx$.

$P = z^2 - 2yz - y^2$, $Q = xy + zx$, $R = xy - zx$.

AE: $\frac{dx}{(z-y)^2} = \frac{dy}{x(y+z)} = \frac{dz}{x(y-z)}$

$\frac{dy}{x(y+z)} = \frac{dz}{x(y-z)} \Rightarrow (y-z)dy = (y+z)dz$

$\Rightarrow ydy - zdy = ydz + zdz$

$\Rightarrow ydy - zdz = ydz + zdy$

$\Rightarrow y^2 - z^2 - 2yz = C$

Choose $l = x$, $m = y$, $n = z$

$x(z^2 - 2yz - y^2) + y(xy + zx) + z(xy - zx)$

$= xz^2 - 2xyz - xy^2 + xy^2 + xyz + xyz - z^2x$

$= xz^2 - 2xyz - xy^2 + xy^2 + 2xyz - xz^2$

$= 0$

then $x dx + y dy + z dz = 0$

$\Rightarrow x^2 + y^2 + z^2 = 0$

Soln. is $f(y^2 - z^2 - 2yz, x^2 + y^2 + z^2) = 0$.

10. Solve $(x^2 - y^2 - z^2)p + 2xyq = 2xz$.

*** $P = x^2 - y^2 - z^2$, $Q = 2xy$, $R = 2xz$.

AE: $\frac{dx}{x^2 - y^2 - z^2} = \frac{dy}{2xy} = \frac{dz}{2xz} = \frac{l dx + m dy + n dz}{lP + mQ + nR}$

Consider $\frac{dy}{2xy} = \frac{dz}{2xz} \Rightarrow \frac{dy}{y} = \frac{dz}{z}$

$$\Rightarrow \log y = \log z + \log c_1$$

$$\Rightarrow \frac{y}{z} = c_1$$

Let $l = x$, $m = y$, $n = z$.

$$\therefore \frac{dx}{x^2 - y^2 - z^2} = \frac{dy}{2xy} = \frac{dz}{2xz} = \frac{x dx + y dy + z dz}{x^3 - xy^2 - xz^2 + 2xy^2 + 2xz^2}$$

$$\Rightarrow \frac{dx}{x^2 - y^2 - z^2} = \frac{dy}{2xy} = \frac{dz}{2xz} = \frac{x dx + y dy + z dz}{x^3 + xy^2 + xz^2}$$

Now, $\frac{dy}{2xy}$ (or $\frac{dz}{2xz}$) $= \frac{x dx + y dy + z dz}{x(x^2 + y^2 + z^2)}$

$$\Rightarrow \frac{dy}{y} = \frac{2(x dx + y dy + z dz)}{x^2 + y^2 + z^2}$$

$$\Rightarrow \log y = \log(x^2 + y^2 + z^2) + \log c_2$$

$$\Rightarrow \frac{y}{x^2+y^2+z^2} = C_2.$$

$$\therefore \text{Solution is } f\left(\frac{y}{z}, \frac{y}{x^2+y^2+z^2}\right) = 0.$$

$$\text{If } u = x^2 + y^2 + z^2.$$

$$du = 2x dx + 2y dy + 2z dz.$$

$$\therefore \int \frac{2x dx + 2y dy + 2z dz}{x^2 + y^2 + z^2} = \int \frac{du}{u} = \log u$$

$$= \log(x^2 + y^2 + z^2).$$

11. Solve $(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$.

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$$P = x^2 - yz, \quad Q = y^2 - zx, \quad R = z^2 - xy.$$

$$\text{AE: } \frac{dx}{x^2 - yz} = \frac{dy}{y^2 - zx} = \frac{dz}{z^2 - xy} = k.$$

$$\frac{dx - dy}{x^2 - yz - y^2 + zx} = \frac{dy - dz}{y^2 - zx - z^2 + xy}$$

$$\Rightarrow \frac{dx - dy}{x^2 - y^2 + z(x - y)} = \frac{dy - dz}{y^2 - z^2 + x(y - z)}$$

$$\Rightarrow \frac{dx - dy}{(x - y)[(x + y) + z]} = \frac{dy - dz}{(y - z)[(y + z) + x]}$$

$$\Rightarrow \frac{dx - dy}{x - y} = \frac{dy - dz}{y - z}$$

$$\Rightarrow \frac{(x - y)}{(y - z)} = C_1$$

$$\begin{aligned} a^3 + b^3 + c^3 - 3abc &= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) \end{aligned}$$

$$\frac{dx + dy + dz}{x^2 - yz + y^2 - zx + z^2 - xy} = \frac{x dx + y dy + z dz}{x^3 - xyz + y^3 - xyz + z^3 - xyz}$$

$$\Rightarrow \frac{dx + dy + dz}{(x^2 + y^2 + z^2 - xy - yz - zx)} = \frac{x dx + y dy + z dz}{x^3 + y^3 + z^3 - 3xyz}$$

$$\Rightarrow \frac{dx + dy + dz}{(x^2 + y^2 + z^2 - xy - yz - zx)} = \frac{x dx + y dy + z dz}{(x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)}$$

$$\Rightarrow (x + y + z)(dx + dy + dz) = x dx + y dy + z dz$$

$$\Rightarrow \cancel{x dx} + x dy + x dz + y dx + \cancel{y dy} + y dz + z dx + \cancel{z dz} = \cancel{x dx} + \cancel{y dy} + \cancel{z dz}$$

$$\Rightarrow x dy + y dx + x dz + z dx + y dz + z dy = 0$$

$$\Rightarrow xy + yz + xz = c_2$$

$$\therefore \text{Solution is } f\left(\frac{x-y}{y-z}, xy+yz+xz\right) = 0.$$

12. Solve $(y+z)p + (z+x)q = x+y$.

 $P = y+z, \quad Q = z+x, \quad R = x+y$

AE: $\frac{dx}{y+z} = \frac{dy}{z+x} = \frac{dz}{x+y} = \frac{ldx + mdy + ndz}{lP + mQ + nR}$

$$\frac{dx - dy}{y - x} = \frac{dy - dz}{z - y}$$

$$\Rightarrow \frac{dx - dy}{x - y} = \frac{dy - dz}{y - z}$$

$$\Rightarrow \frac{x-y}{y-z} = c_1$$

Now, $\frac{dx - dy}{-(x-y)} = \frac{dx + dy + dz}{2(x+y+z)}$

$$\Rightarrow -\log(x-y) + \log c_2 = \frac{1}{2} \log(x+y+z)$$

$$\Rightarrow \log(x+y+z)^{1/2} + \log(x-y) = \log c_2.$$

$$\Rightarrow \frac{(x+y+z)^{1/2}}{(x-y)} = c_2.$$

$$\Rightarrow \frac{(x-y)^2}{(x+y+z)} = c_2.$$

$$\therefore \text{Solution is } f\left(\frac{x-y}{y-z}, \frac{(x-y)^2}{(x+y+z)}\right) = 0.$$

$$\textcircled{1} \text{ --- } 0 = p + q + pq \quad \text{Solve}$$

$$0 = (p, q) \text{ is an eqn. of the form } p^2 + q^2 = 0.$$

$$\text{let } q = a$$

$$0 = p + a + pa \quad \Leftrightarrow \textcircled{1}$$

$$a = p(1+a) \quad \Leftrightarrow$$

$$\frac{a}{1+a} = p \quad \Leftrightarrow$$

$$d + p \frac{a}{1+a} + xa = z \quad \text{is } \textcircled{1} \text{ is satisfied } \therefore$$